Understanding of Traps Causing Random Telegraph Noise Based on Experimentally Extracted Time Constants and Amplitude

Kenichi Abe[1], Akinobu Teramoto[2], Shigetoshi Sugawa[1], and Tadahiro Ohmi[2, 3]

1 Graduate School of Engineering

2 New Industry Creation Hatchery Center

3 World Premier International Research Center

Tohoku University

6-6-10, Aza-Aoba, Aramaki, Aoba-ku, Sendai, Japan, 980-8579

Phone: +81-22-795-3977, teramoto@fff.niche.tohoku.ac.jp

***Abstract*— We develop a high-speed method to extract time constants and noise amplitude of random telegraph noise (RTN). We investigate distributions of these RTN parameters for more than 270 n- and p-MOSFETs and clarify spectroscopy of traps causing RTN. Most of traps are distributed in an energy range of 220 meV, and mean times to capture/emission are measured in a wide range between 10 μs and 20 ms.**

Keywords-random telegraph noise, trap, time constant, energy distribution

I. INTRODUCTION

Random telegraph noise (RTN) of metal oxide semiconductor field effect transistors (MOSFETs) has been investigated energetically in recent years because it has a singular characteristic which relates to single electron trapping/detrapping physics and it gives us beneficial information on a trap spectroscopy of the gate dielectric film.

Moreover, RTN has a negative impact on reliable operations of flash memory [1] and static random access memory (SRAM) [2, 3]. The traps causing RTN is also suspected as a cause of negative bias temperature instability (NBTI) in MOSFETs with a high-k/Metal Gate (MG)-gate stack [4]. Therefore, the importance of RTN analysis has become greater.

The transition process of a carrier between the channel region and the trap is still unclear. We focus on time constants of RTN, which represent trapping/detrapping probability and we measure them for numerous samples by a newly developed method to describe the trap spectroscopy experimentally. We have reported that the energy distribution of traps in the gate insulator can be evaluated by an extracting only the ratio of time constants in our newly developed test pattern [5]. However, the time constants themselves cannot be extracted by the previous method because of the long data sampling period. In this paper, very short sampling period of 1 μs was realized and this allows us to understand relationships among RTN parameters which include its amplitude, transition rate, and time constants themselves and physical description of RTN origins. Moreover, samples with different RTN characteristics are investigated by the same way and we clarify the differences precisely.

II. ANALYTICAL APPROACH

We employ an arrayed device under test (DUT) test pattern previously reported by [6] and a newly measurement system to measure numerous transistors (Fig. 1). In this system, RTN waveform appears as source voltage fluctuation at each cell under applied constant drain current.

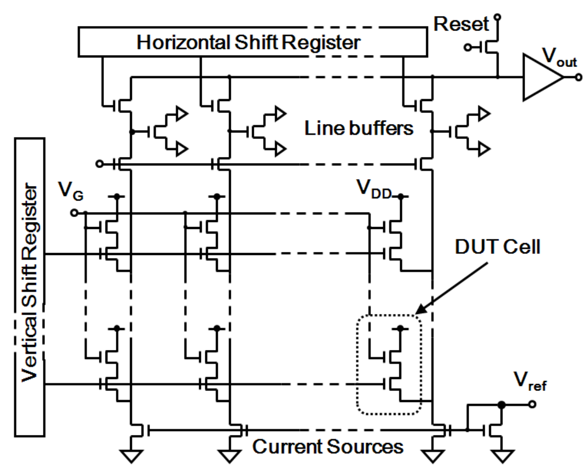

Figure 1. Schematic of the test circuit. The chip includes 131,072 nFETs and 81,920 p-MOSFETs as DUT transistors.

Two-level type RTN is characterized by only three parameters, which are mean time to capture ($<\tau_c>$), mean time to emission ($<\tau_e>$), and amplitude (ΔV_{gs}). The time constants correspond to two physical states of a trap, that is τ_c and τ_e represent spans in low V_{gs} level (carrier trapping state) and those in high V_{gs} level (carrier emission state) (Fig. 2(a)). The RTN amplitude ΔV_{gs} is defined as a difference between two normal distributions in a voltage histogram (Fig. 2(b)).

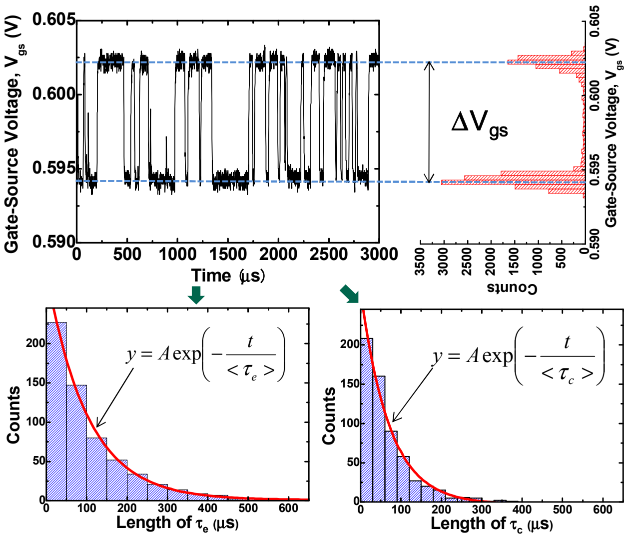

Figure 2. Definitions and extractions of two time constants ($<\tau_c>$, $<\tau_e>$) and amplitude (ΔV_{gs}) from RTN waveform data.

To extract time constants accurately, we select a particular cell showing RTN by pseudo-addressing with two shift registers and digitize the noise signal by an oscilloscope with fast sampling rate and long record length. A determination of RTN existence uses two criteria, which consists of over 1.7 mV of ΔV_{gs} due to limitation of the instruments accuracies and over 80 jumps between high- and low-level. In this work, we measured the test circuit under 1 MHz of the sampling rate and 10^6 points (1 s) of the record length, respectively. We extract the time constants by fitting the distributions of τ_c and τ_e to the exponential distribution ($Ae^{-t/<\tau>}$) because the phenomenon is ruled by Poisson process.

The test chip includes 131,072 n-MOSFETs and 81,920 p-MOSFETs with 0.22 µm of gate length, 0.28 µm of gate width, and 5.7 nm wet thermal gate oxide. It was fabricated by 0.22 µm logic CMOS technology.

III. Results and Discussions

Fig. 3 shows a histogram of E_t-E_{fn} where E_t is the trap energy, E_{fn} is quasi-Fermi energy of electron. We can calculate the trap energy E_t from τ_c and τ_e by a following equation,

$$\frac{\langle\tau_c\rangle}{\langle\tau_e\rangle} = g\cdot\exp\left\{\left(\frac{E_t - E_{fn}}{kT}\right)\right\} \quad (1)$$

where g is the degeneracy, k is Boltzmann constant, and T is the temperature [7]. In this graph, the distribution has a symmetrical form at the point of $E_t = E_{fn}$ and E_t-E_{fn} spreads to about ±110 meV. We have reported the histogram E_t-E_{fn} for a similar measurement condition (I_d=0.1 µA) and different record period [5]. Fig. 4 shows the histogram E_t-E_{fn} reported in [5]. The record period of 1 s in this experiment is much shorter than that of 4,200 s in [5]. The distributions of the detected trap energy are different between two conditions. Fig. 5 shows the trap energy distributions with the band diagram of the silicon channel region at the source edge for both record periods. The bottom energy of the conduction band at in the channel at the source edge is 12.3mV, which is simulated by the device simulator (SILVACO, Inc. ATLAS). The sub-band energy E_0 (Lowest energy) and E_1 (second lowest energy) are calculated by following equation [8].

$$E_j = \left\{\frac{3hq\varepsilon_S}{4\sqrt{2m_x}}\left(j+\frac{3}{4}\right)\right\}^{\frac{2}{3}},\ j=1,2,3\cdots \quad (2)$$

Where, ε_s is electric field. E_j is j-th's sub-band energy, h and m_x are Planck's constant and effective mass of electrons, respectively. The E_0 and E_1 are 52.3 and 82.8 mV, respectively. The detected traps at only higher energy than E_{fn} increase with the sampling period though those at lower energy than E_{fn} do not increases. These suggest that the trap having a higher energy than E_{fn} is hard to detect in short record period because the high energy carrier density in the channel region is smaller than that near E_{fn}, and the number of traps located at lower energy level than E_{fn} are less than that at higher energy.

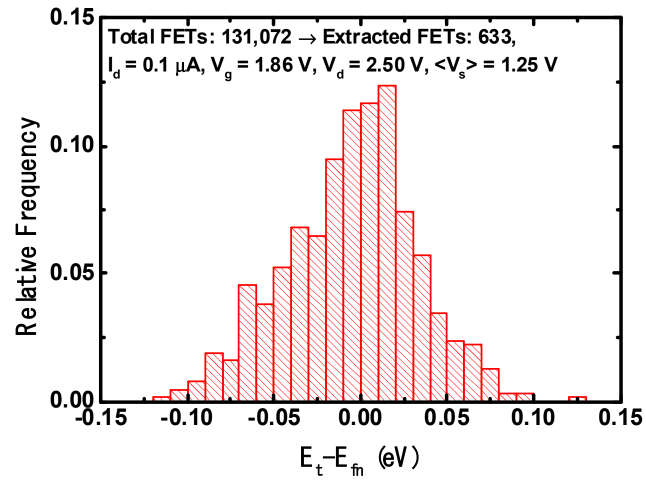

Figure 3. Histogram of E_t-E_{fn} for n-MOSFETs calculated from $<\tau_c>/<\tau_e>$. Sampling cycle=1 µs, Record Period=1 s.

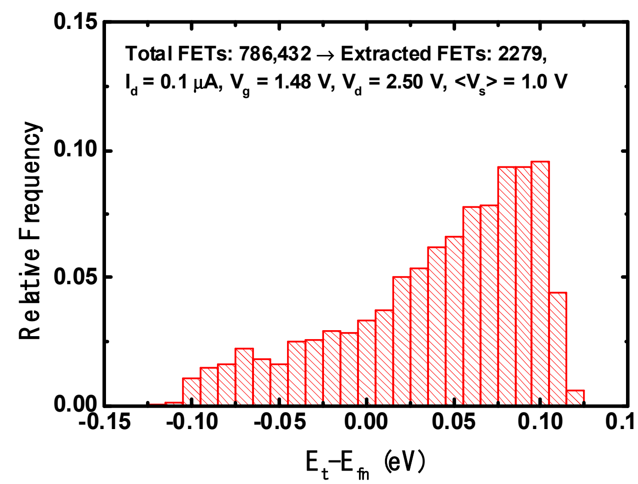

Figure 4. Histogram of E_t-E_{fn} for n-MOSFETs calculated from $<\tau_c>/<\tau_e>$. Sampling cycle=0.7 s, Record Period=4,200 s [5].

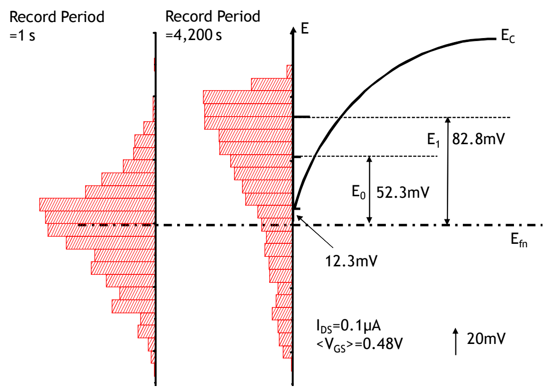

Figure 5. Trap energy distributions with the band diagram of the silicon channel region for the measurement record periods of 1 s and 4,200 s.

Fig. 6 show distributions of $<\tau_c>$ and $<\tau_e>$ for n-MOSFETs. Both of them distribute based on a power function in the wide range of 10 µs to 20 ms because of the variability of the trap energy and position. In these graph, the vertical axis also corresponds to a trap density $N_t(E_t, x_t)$ where x_t is trap depth from the interface between Si and the gate dielectric. In our experiments, numerous traps where are near the interface and which have small E_t-E_{fn} are observed and they may affect actual circuit operations and reliabilities.

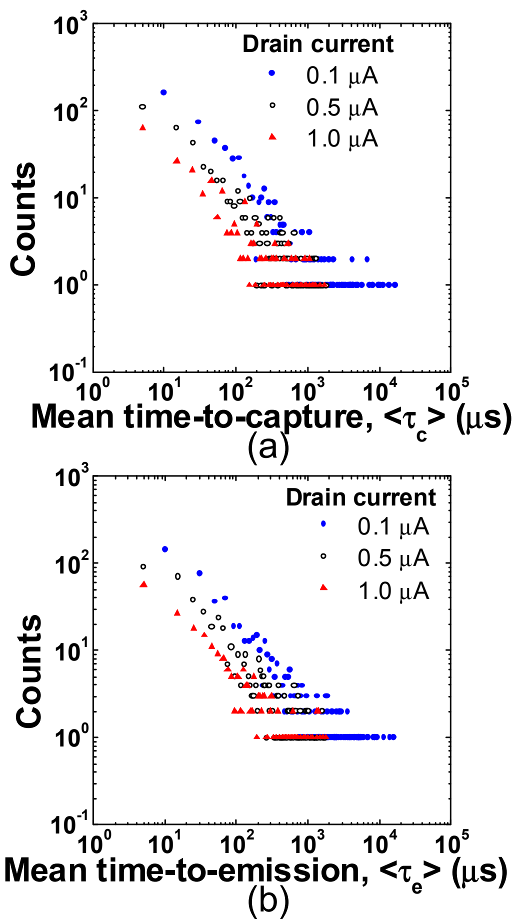

Figure 6. Distributions of time constants of (a) $<\tau_c>$ and (b) $<\tau_e>$ for three drain currents.

These experimental results can be used for a prediction of occurrence frequency of problematic time constants in an actual circuit design such as SRAM [9]. Furthermore, the distributions move to shorter as drain current increases. It can be explained by a simple relation,

$$\tau_c = \frac{1}{n\sigma v_{th}} \tag{3}$$

where n is the carrier density in the inversion layer, σ is the capture cross section, and v_{th} is the average thermal velocity of the carriers [7]. Increase in the drain current means increase in the carrier density (n) and it results in decrease in τ_c and increase in the transition rate averagely. Therefore, the relationship between τ_c and τ_e shows positive correlation (Fig. 7). This means that the trapping and detrapping process are not independent of each other. In other words, they almost occur between the inversion layer and the trap site, the case of detrapping to the gate electrode rarely occurs.

Fig. 8 show scatter plots between number of jumps N_J and $<\tau_c>$, $<\tau_e>$. Negative correlation appears because the inverses of time constants represent the transition rate. Fig. 9 shows the relationship between the number of jumps N_J and E_t-E_{fn}. Here, the relation of the number of jumps, the sampling times, and the trap energy level is as follows [5, 7, 10]

$$\frac{\langle\tau_c\rangle}{\langle\tau_e\rangle} = g\cdot\exp\left\{\left(\frac{E_t - E_f}{kT}\right)\right\} = \frac{Count\_L}{Count\_H} \tag{4}$$

Where Count_L and Count_H are the data count number of the low (τ_c period) and high (τ_e period) states, respectively. In this experiment, the total sampling number is 10^6. Then,

$$Count\_L + Count\_H = 10^6 \tag{5}$$

From (3) and (4),

$$Count\_L = \frac{10^6 \times A}{1 + A} \tag{6}$$

$$Count\_H = \frac{10^6}{1 + A} \tag{7}$$

Where,

$$A = g\cdot\exp\left(\frac{E_t - E_f}{kT}\right) \tag{8}$$

The maximum number of jumps N_{JMAX} is indicated by (9).

$$N_{JMAX} = 2\times\min(Count\_L, Count\_H) \tag{9}$$

As indicating (6)-(9), N_{JMAX} has to depend on the trap energy level. The shape of maximum value of N_J in Fig.9 is defined by these reasons. N_J is widely distributed in each trap energy level. It is considered that this distribution indicates the distance between traps and channels. The large and small N_J's mean the near and far between traps and channels, respectively. On the other hand, the minimum number of jumps was 80, then, the minimum Count_(L or H) becomes 40. From (3), the detectable E_t-E_f range become about ±200 meV. In this experiment, E_t-E_f distribution was in ±110 meV. This indicates that the E_t-E_f of detected traps in this experiment was ±110 meV.

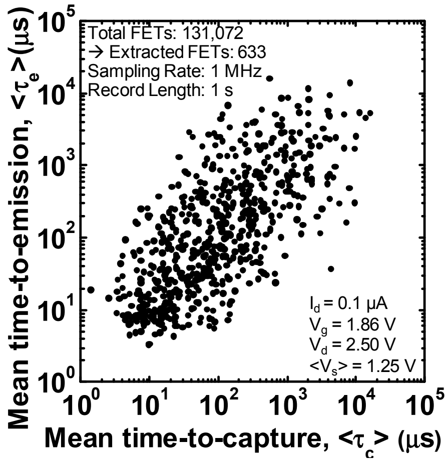

Figure 7. Scatter plot between $\langle\tau_c\rangle$ and $\langle\tau_e\rangle$. Positive correlation appears within the range of 10 μs to 10 ms.

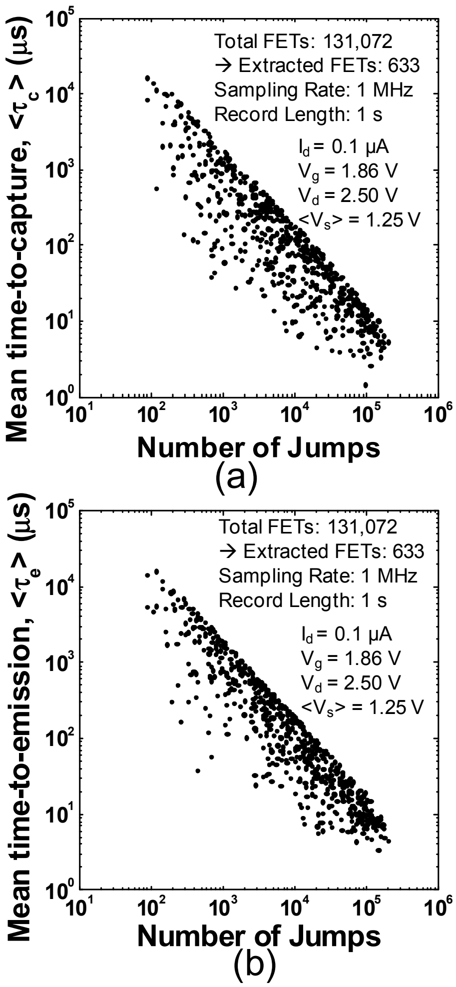

Figure 8. Scatter plots between number of jumps and (a) $\langle\tau_c\rangle$, (b) $\langle\tau_e\rangle$.

Fig. 10 show scatter plots between ΔV_{gs} and $\langle\tau_c\rangle$, $\langle\tau_e\rangle$. There are not clear correlations. It seems that the trap locating on near the Si/SiO_2 interface causes shorter $\langle\tau_c\rangle/\langle\tau_e\rangle$ and larger ΔV_{gs} simultaneously, however ΔV_{gs} is mainly dominated not by the trap depth from the interface, but by the randomness of the channel percolation and the lateral trap location [11, 12].

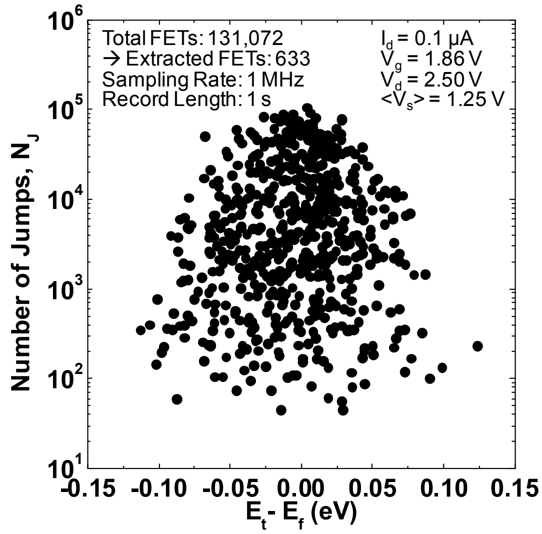

Figure 9. Relationship between the number of jumps and E_t-E_{fn}.

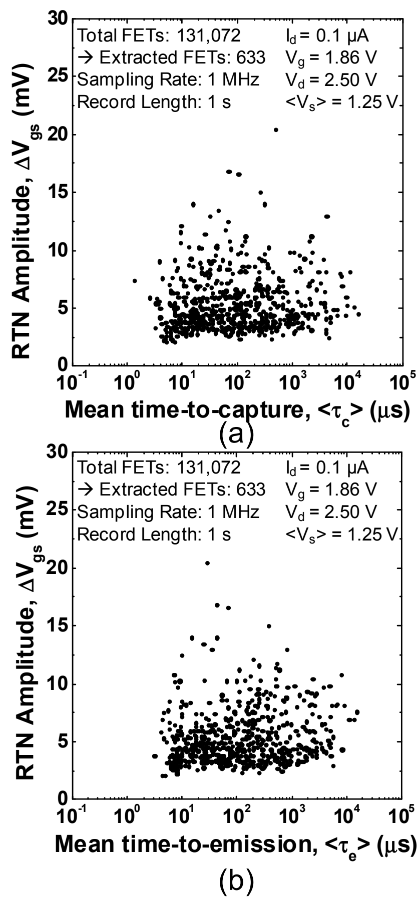

Figure 10. Scatter plots between number of jumps and (a) $\langle\tau_c\rangle$, (b) $\langle\tau_e\rangle$.

We can also investigate RTN for p-MOSFETs with the same system. Figure 11 indicate relative frequency distributions of the time constants for n- and p-MOSFETs. Slopes of p-MOSFET in the double logarithmic plots are smaller for $<\tau_c>$ and $<\tau_e>$ than ones of n-MOSFETs. Possible causes are considered such as differences in the trap distribution, the carrier density, and the capture cross section for electron and hole. Fig. 12 shows histograms of E_t-E_f for n- and p-MOSFETs where E_f means quasi-Fermi energy of electron and hole, respectively. A slight difference in the symmetric property can be caused by the differences in the trap distributions for electrons and holes.

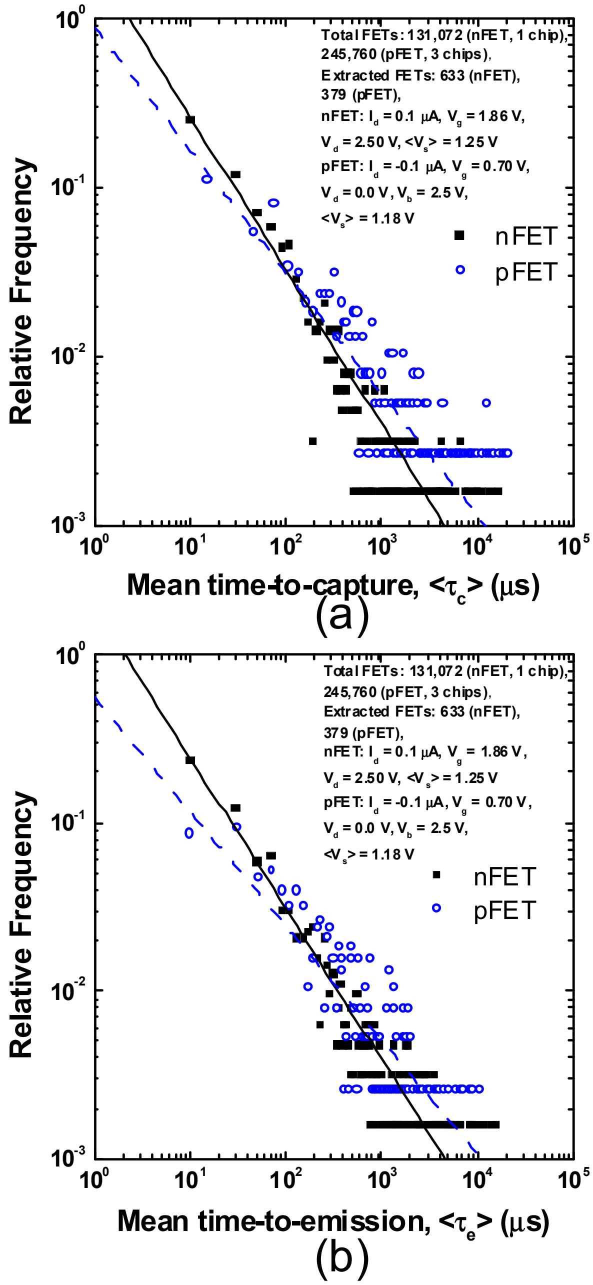

Figure 11. Relative frequency distributions of time constants of (a) $<\tau_c>$ and (b) $<\tau_e>$ for n- and p-MOSFETs.

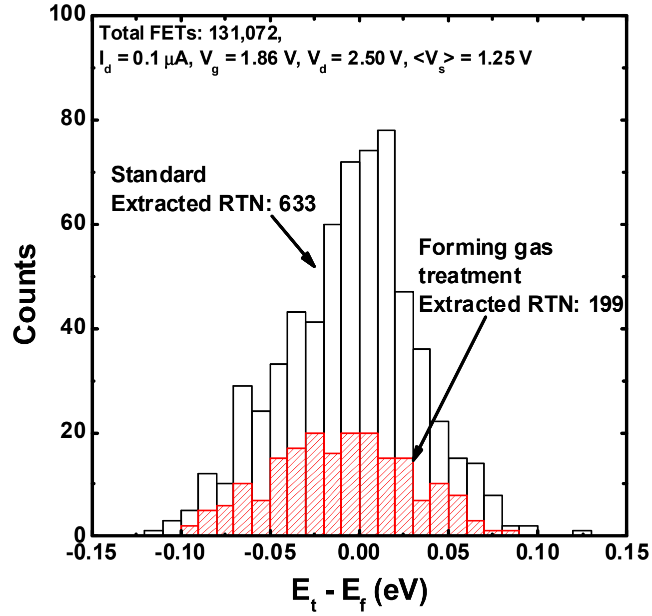

Figure 12. Comparison of histograms of E_t-E_f for n- and p-MOSFETs.

IV. CONCLUSION

We have clarified the trap spectroscopy relating to RTN from the viewpoint of $<\tau_c>$ and $<\tau_e>$ for numerous MOSFETs experimentally. The result of the positive correlation between $<\tau_c>$ and $<\tau_e>$ shows that trapping and detrapping process are same fundamentally. The amplitude ΔV_{gs} has no relation to $<\tau_c>$ and $<\tau_e>$, and ΔV_{gs} is almost ruled by randomness of trap location and channel percolation. RTN time constants of n-MOSFET and p-MOSFET are slightly different but the trap density of p-MOSFET is very small than that of n-MOSFET. With this method, RTN trap spectroscopy will be measured in a short time and we can easily obtain important information on circuit operation margins, reliabilities, and physical mechanism of RTN.

ACKNOWLEDGMENT

We would like to thank Mr. Y. Kamata and Mr. K. Shibusawa for useful discussions and manufacture of the test structure.

REFERENCES

[1] N. Tega, H. Miki, T. Osabe, A. Kotabe, K. Otsuga, H. Kurata, S. Kamohara, K. Tokami, Y. Ikeda, and R. Yamada, "Anomalously Large Threshold Voltage Fluctuation by Complex Random Telegraph Signal in Floating Gate Flash Memory," in *International Electron Devices Meeting*, San Francisco, 2006, pp. 491-494.

[2] S. O. Toh, Y. Tsukamoto, G. Zheng, L. Jones, L. Tsu-Jae King, and B. Nikolic, "Impact of random telegraph signals on V_{min} in 45nm SRAM," in *IEEE International Electron Devices Meeting*, Baltimore, 2009, pp. 767-770.

[3] M. Tanizawa, S. Ohbayashi, T. Okagaki, K. Sonoda, K. Eikyu, Y. Hirano, K. Ishikawa, O. Tsuchiya, and Y. Inoue, "Application of a Statistical Compact Model for Random Telegraph Noise to Scaled-

SRAM Vmin Analysis," in *Digest of Technical Papers Symposium on VLSI Technology*, Honolulu, 2010, pp. 95-96.

[4] H. Reisinger, T. Grasser, W. Gustin, Schlu, x, and C. nder, "The statistical analysis of individual defects constituting NBTI and its implications for modeling DC- and AC-stress," in *2010 IEEE International Reliability Physics Symposium* Anaheim, 2010, pp. 7-15.

[5] A. Teramoto, T. Fujisawa, K. Abe, S. Sugawa, and T. Ohmi, "Statistical evaluation for trap energy level of RTS characteristics," in *Symposium on VLSI Technology* Honolulu, 2010, pp. 99-100.

[6] K. Abe, S. Sugawa, S. Watabe, N. Miyamoto, A. Teramoto, Y. Kamata, K. Shibusawa, M. Toita, and T. Ohmi, "Random Telegraph Signal Statistical Analysis using a Very Large-scale Array TEG with 1M MOSFETs," in *IEEE Symposium on VLSI Technology*, Kyoto, 2007, pp. 210-211.

[7] M. J. Kirton and M. J. Uren, "Noise in solid-state microstructures: A new perspective on individual defects, interface states and low-frequency (1/f) noise," *Advances in Physics,* vol. 38, pp. 367 - 468, 1989.

[8] Y. Taur and T. H. Ning, "Fundamentals of Modern VLSI Devices," 2nd ed Cambridge: Cambridge University Press, 2009, pp. 234-239.

[9] K. Takeuchi, T. Nagumo, K. Takeda, S. Asayama, S. Yokogawa, K. Imai, and Y. Hayashi, "Direct observation of RTN-induced SRAM failure by accelerated testing and its application to product reliability assessment," in *Symposium on VLSI Technology*, 2010, pp. 189-190.

[10] T. Fujisawa, K. Abe, S. Watabe, N. Miyamoto, A. Teramoto, S. Sugawa, and T. Ohmi, "Accurate Time Constant of Random Telegraph Signal Extracted by a Sufficient Long Time Measurement in Very Large-Scale Array TEG," in *IEEE International Conference on Microelectronic Test Structures*, 2009, pp. 19-24.

[11] K. Abe, A. Teramoto, S. Watabe, T. Fujisawa, S. Sugawa, Y. Kamata, K. Shibusawa, and T. Ohmi, "Experimental Investigation of Effect of Channel Doping Concentration on Random Telegraph Signal Noise," *Jpn. J. Appl. Phys.,* vol. 49, p. 04DC07, 2010.

[12] A. Ghetti, C. M. Compagnoni, A. S. Spinelli, and A. Visconti, "Comprehensive Analysis of Random Telegraph Noise Instability and Its Scaling in Deca-Nanometer Flash Memories," *IEEE Trans Electron Devices,* vol. 56, pp. 1746-1752, Aug 2009.